SPI Slave + Single-Port RAM – RTL Specification

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1. Introduction

This document specifies an SPI slave that gives a master read and write access to a 256 x 8 single-port RAM. Every behaviour is written as a numbered requirement so it can be extracted as a feature in section 9 and verified.

Scope. The spec covers three RTL modules. It does not cover the SPI master, pad cells, tri-state control of MISO, multi-slave buses, DFT or power.

Module	File	Role
`spi_wrapper`	`rtl/spi_wrapper.v`	Top level; wires the slave to the RAM
`spi_slave`	`rtl/spi_slave.v`	SPI mode 0 front end: receives 10-bit frames, shifts out read data
`spi_ram`	`rtl/spi_ram.v`	Command decoder + single-port memory

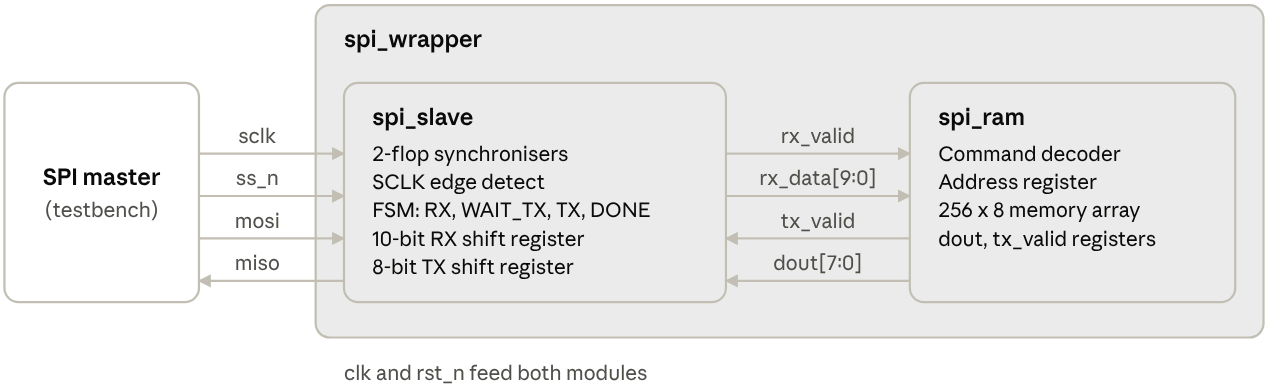
Conventions.

- Requirement IDs use the prefixes `SPI-` (slave), `RAM-` (memory) and `TOP-` (wrapper and system level).
- "Shall" marks a testable requirement; each one maps to at least one feature ID (`F-nn`) in section 9.
- Times are in `clk` cycles unless stated. Reference testbench values: `clk` = 10 ns, SCLK period = 200 ns.
- Bit ranges follow Verilog order, for example `rx_data[9:8]` is the command.

2. Architecture overview

A master sends 10-bit command frames over SPI. `spi_slave` turns each frame into a one-cycle `rx_valid` pulse. `spi_ram` executes the command and, for a read, hands the byte back for shifting out on MISO.

The slave turns SPI frames into RAM commands and returns read data



block diagram · 3 modules, 6 pins, 4 handshake signals

The slave owns all SPI timing and the RAM owns all memory state, so the two can be verified separately and joined through the four handshake signals.

- **`spi_slave`**: synchronises the SPI pins, counts bits, assembles the 10-bit frame and shifts read data out.
- **`spi_ram`**: decodes `rx_data[9:8]`, holds the address register and the memory, and returns read data with `tx_valid`.
- **`spi_wrapper`**: wiring only, no logic.

Write. Frame `00` sets the address; frame `01` writes `byte` to `mem[addr]`. Nothing is returned on MISO.

Read. Frame `10` sets the address. Frame `11` makes the RAM load `dout` from `mem[addr]` and pulse `tx_valid`. The slave latches the byte and shifts it out on the next 8 falling SCLK edges.

3. Interfaces

The wrapper exposes six pins; the slave and RAM talk over four internal handshake signals. All signals are single-bit unless a width is shown.

`spi_wrapper` ports

Port	Dir	Width	Description
`clk`	in	1	System clock, rising edge active

Port	Dir	Width	Description
`rst_n`	in	1	Asynchronous reset, active low
`sclk`	in	1	SPI clock, idle low, asynchronous to `clk`
`ss_n`	in	1	Slave select, active low, idle high
`mosi`	in	1	Master out, slave in
`miso`	out	1	Master in, slave out; always driven, 0 when idle

Slave to RAM handshake

Signal	Driver	Width	Description
`rx_valid`	`spi_slave`	1	One-cycle pulse: a full 10-bit frame was received
`rx_data`	`spi_slave`	10	`{cmd[1:0], byte[7:0]}`; valid while `rx_valid` is high
`tx_valid`	`spi_ram`	1	One-cycle pulse: `dout` holds the requested read byte
`tx_data` (RAM `dout`)	`spi_ram`	8	Byte for the slave to shift out on MISO

`spi_ram` parameters

Parameter	Default	Constraint
`DEPTH`	256	Shall equal 2^`ADDR_W`
`ADDR_W`	8	At most 8, because the address travels in one byte

`spi_wrapper` has no parameters, so DEPTH and ADDR_W cannot be changed from the top level without editing the wrapper.

4. SPI protocol

The slave uses SPI mode 0 (CPOL=0, CPHA=0), MSB first. The master samples MISO on rising SCLK edges. The slave samples MOSI on rising edges and changes MISO on falling edges. Each transaction is one frame inside one SS_n assertion.

Frame format. A command frame is 10 bits: `cmd[1:0]` first, then `byte[7:0]`.

`cmd`	Name	`byte` meaning	Effect
`00`	Write address	Address	Loads the RAM address register
`01`	Write data	Data	Writes `byte` to `mem[addr]`
`10`	Read address	Address	Loads the RAM address register
`11`	Read data	Ignored (dummy)	Slave then shifts out `mem[addr]` on MISO

Write sequence (two frames, SS_n returns high between them):

1. SS_n low, frame `00` + address (10 SCLK pulses), SS_n high.
2. SS_n low, frame `01` + data (10 SCLK pulses), SS_n high.

Read sequence (two frames):

1. SS_n low, frame `10` + address (10 SCLK pulses), SS_n high.
2. SS_n low, frame `11` + dummy byte (10 SCLK pulses), then 8 more SCLK pulses while the master samples MISO, SS_n high.

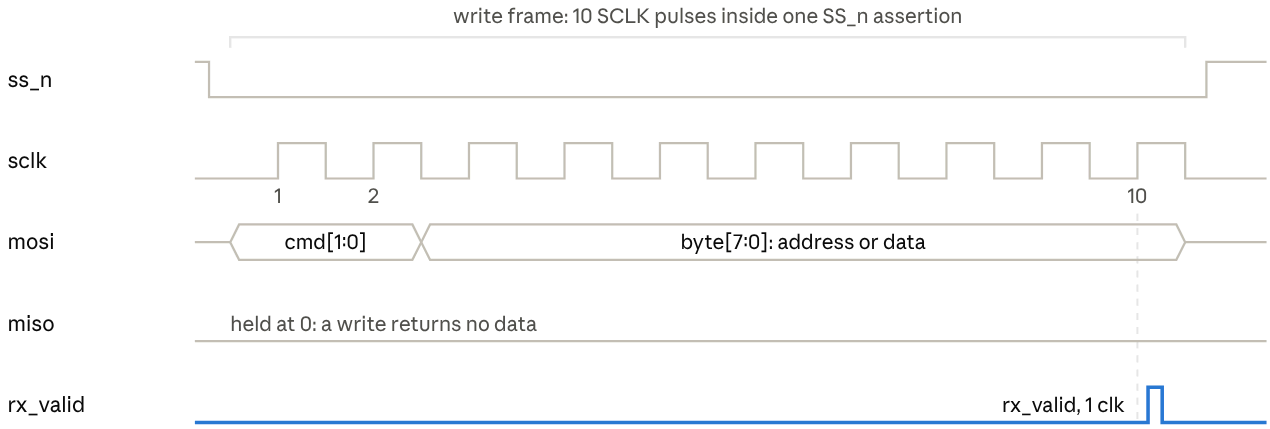
Read frame, pulse by pulse

SCLK pulse	MOSI	MISO
1 to 10	`1`, `1`, then 8 dummy bits	0
Falling edge of pulse 10	Don't care	Slave drives `tx_data[7]` a few `clk` cycles later
11 to 18	Don't care	Bits 7 down to 0, sampled by the master on each rising edge

The address register is not incremented by any command. Each write or read therefore needs its own address frame. After the last bit the slave holds MISO at bit 0 until SS_n goes high.

Timing diagrams

A write frame is 10 SCLK pulses, and rx_valid follows the 10th rising edge

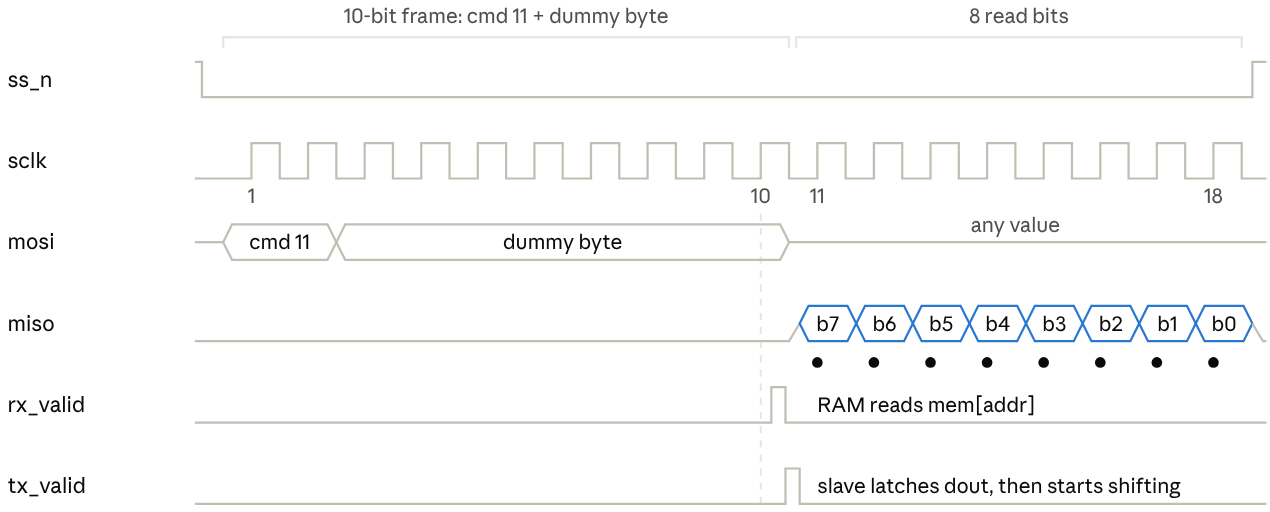


On the next clk the RAM acts: cmd 00 or 10 loads addr, cmd 01 writes mem[addr].
Not to scale: clk cycles are exaggerated. The slave samples MOSI on each rising edge.

write frame timing · 10 SCLK pulses

SS_n falls before the first pulse. The master changes MOSI while SCLK is low and the slave samples it on each rising edge. The 10th rising edge completes the frame, and `rx_valid` follows a few `clk` cycles later. A write frame leaves MISO at 0.

In a read, MISO changes on falling edges and is sampled on rising edges



Dots mark the rising edges where the master samples MISO.
Not to scale: clk cycles are exaggerated.

read frame timing · 18 SCLK pulses, 8 MISO bits

The command frame is identical to a write. After the 10th falling edge the slave drives bit 7 on MISO, and the master samples bits 7 down to 0 on rising edges 11 to 18. MISO holds bit 0 until SS_n goes high.

5. Functional requirements

There are 26 requirements: 12 for the slave, 10 for the RAM and 4 for the top level. Each one is a single testable statement.

SPI slave

ID	Requirement
SPI-001	The slave shall sample MOSI on each rising SCLK edge, MSB first (mode 0).
SPI-002	A command frame shall be 10 bits: `cmd[1:0]` followed by `byte[7:0]`.
SPI-003	After the 10th rising SCLK edge the slave shall pulse `rx_valid` for exactly one `clk` cycle and present the 10-bit frame on `rx_data`.

ID	Requirement
SPI-004	While SS_n is high the slave shall reset its bit counters, return to the receive state and drive MISO to 0.
SPI-005	For `cmd` 11 the slave shall wait for `tx_valid`, then latch `tx_data`.
SPI-006	The slave shall change MISO only on falling SCLK edges, starting with `tx_data[7]` on the 10th falling edge, and shall shift out 8 bits MSB first.
SPI-007	After a frame with `cmd` 00, 01 or 10 the slave shall ignore SCLK until SS_n goes high.
SPI-008	After the 8th read bit the slave shall hold MISO at `tx_data[0]` and ignore further SCLK pulses until SS_n goes high.
SPI-009	The slave shall pass SCLK, SS_n and MOSI through two-flop synchronisers; all internal logic shall run on `clk`.
SPI-010	Asserting `rst_n` shall asynchronously set the state to receive and clear counters, `rx_valid`, `rx_data` and MISO.
SPI-011	Only one frame shall be decoded per SS_n assertion.
SPI-012	A frame cut short by SS_n going high before the 10th rising edge shall produce no `rx_valid`; the next frame shall start counting at bit 1.

RAM

ID	Requirement
RAM-001	Reset shall clear the address register, `dout` and `tx_valid`. Reset shall not clear memory contents.
RAM-002	On `rx_valid` with `cmd` 00 the RAM shall load the address register from `din[ADDR_W-1:0]`.

ID	Requirement
RAM-003	On `rx_valid` with `cmd` 01 the RAM shall write `din[7:0]` to `mem[addr]` and leave `addr` unchanged.
RAM-004	On `rx_valid` with `cmd` 10 the RAM shall load the address register from `din[ADDR_W-1:0]`, with the same effect as `cmd` 00.
RAM-005	On `rx_valid` with `cmd` 11 the RAM shall load `dout` from `mem[addr]` and pulse `tx_valid` in the next cycle. `din[7:0]`, `addr` and memory shall be unaffected.
RAM-006	When `rx_valid` is low, the RAM shall change no state except returning `tx_valid` to 0.
RAM-007	`tx_valid` shall be high for exactly one cycle, and only in response to `cmd` 11.
RAM-008	The RAM shall perform at most one memory access per `clk` cycle.
RAM-009	Capacity shall be `DEPTH` x 8 bits, addressed 0 to `DEPTH`-1.
RAM-010	In simulation every location shall read 0x00 until written (initial block; not guaranteed after synthesis).

Top level

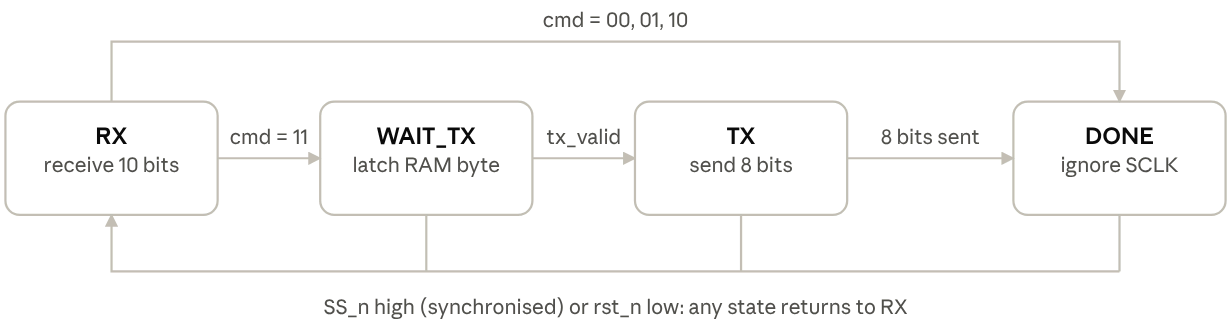
ID	Requirement
TOP-001	`rx_valid` and `rx_data` shall connect from the slave to the RAM; `tx_valid` and `dout` shall connect from the RAM to the slave.
TOP-002	`miso` shall be a plain driven output, 0 when idle, with no tri-state.

ID	Requirement
TOP-003	`clk` shall be at least 8 times faster than SCLK (SCLK half-period of at least 4 `clk` cycles). See section 7.
TOP-004	SCLK shall idle low and SS_n shall idle high.

6. Microarchitecture

The slave is one four-state FSM clocked by `clk`; the RAM is a single registered decoder with no state machine.

Every frame ends in DONE, and SS_n high returns any state to RX



slave FSM · 4 states, 5 transitions

The 10th rising SCLK edge ends RX. The next state depends on the top two bits of the frame: `cmd` 11 goes to WAIT_TX, anything else goes to DONE. The 8th falling edge in TX leads to DONE. While `ss_n` is high the slave sits in RX with both counters at 0.

Registers

Register	Width	Module	Function
`sclk_s`, `ss_s`, `mosi_s`	2 each	`spi_slave`	Synchroniser chains; bit 1 feeds the logic
`sclk_d`	1	`spi_slave`	Previous synchronised SCLK, for rising and falling edge detect
`state`	3	`spi_slave`	RX = 0, WAIT_TX = 1, TX = 2, DONE = 3

Register	Width	Module	Function
`rx_cnt`	4	`spi_slave`	Rising edges seen in the frame, 0 to 9
`rx_shift`	9	`spi_slave`	First 9 received bits; the 10th is appended from `mosi` to form `rx_data`
`tx_shift`	8	`spi_slave`	Read byte, shifted left on each falling edge
`tx_cnt`	3	`spi_slave`	Bits shifted out, 0 to 7
`miso`, `rx_valid`, `rx_data`	1, 1, 10	`spi_slave`	Registered outputs
`addr`	`ADDR_W`	`spi_ram`	Shared read and write address
`mem`	`DEPTH` x 8	`spi_ram`	Memory array
`dout`, `tx_valid`	8, 1	`spi_ram`	Registered read result and its valid pulse

7. Timing, clocking and reset

`clk` must run at least 8 times faster than SCLK. The slave needs about 6 `clk` cycles after the 10th rising SCLK edge to fetch the RAM byte and be ready to shift. It then needs 3 more cycles after a falling edge to update MISO. This ratio comes from the latency analysis below, not from simulation; the 8:1 boundary is a verification item (F-11).

Latencies (clock edges counted from the SCLK pin event)

Event	Latency
SCLK, SS_n or MOSI pin change to internal synchronised value	2 `clk` edges
10th rising SCLK edge to `rx_valid` high	3 `clk` edges
`rx_valid` (`cmd` 11) to RAM `tx_valid` high	1 `clk` edge
`tx_valid` high to slave in transmit state	1 `clk` edge
Falling SCLK edge to MISO change	3 `clk` edges

Why 8:1. The slave is ready to shift 6 edges after the 10th rising edge. The 10th falling edge arrives half an SCLK period later and is acted on 3 edges after that, so the half-period must be at least 3 `clk` cycles. MISO must also settle before the next rising edge, which needs at least 4 cycles. Hence a minimum SCLK period of 8 `clk` cycles. The reference testbench uses 20.

Reset

- `rst_n` is asynchronous and active low on every flop in all three modules.
- Assertion puts the slave in the receive state with `miso`, `rx_valid`, `rx_data` and counters at 0, and the RAM with `addr`, `dout`, `tx_valid` at 0.
- Memory contents are not cleared by reset.
- Deassertion is not synchronised inside the design. The system shall release `rst_n` synchronously to `clk`.

8. Corner cases, assumptions and limitations

The RTL defines the behaviour in the first table; the second lists assumptions and gaps the design does not cover, which the verification plan should either test or flag.

Defined corner-case behaviour

Case	Behaviour	Requirement
SS_n rises after fewer than 10 bits	Frame discarded, no `rx_valid`, next frame starts at bit 1	SPI-012

Case	Behaviour	Requirement
SS_n rises during a read, before 8 bits are out	Slave returns to receive, MISO goes to 0	SPI-004
More than 10 SCLK pulses after a write or address frame	Extra pulses ignored	SPI-007
More than 18 SCLK pulses in a read	Extra pulses ignored, MISO holds bit 0	SPI-008
Second frame inside the same SS_n assertion	Not decoded	SPI-011
Read of an address never written	Returns 0x00 in simulation	RAM-010
Write then read of the same address	Read returns the newest written value	RAM-003, RAM-005
Write data command without a prior address frame	Writes to the current `addr` (0 after reset)	RAM-003
Consecutive reads without a new address frame	Return the same location each time	RAM-005
`cmd` 11 dummy byte takes any value	Ignored	RAM-005
Reset during a transaction	Slave and RAM registers cleared, memory kept	SPI-010, RAM-001

Assumptions and known limitations

- SCLK idles low and SS_n idles high (TOP-004). The slave does not protect against SCLK idling high.
- SS_n should stay high for at least 3 `clk` cycles between frames, and the first rising SCLK edge should come at least half an SCLK period after SS_n falls. The reference master gives much more; these minimums are assumptions to confirm in simulation.
- MISO is not tri-stated, so several slaves cannot share the line.
- There is no address auto-increment and no burst mode.
- Reset deassertion is not synchronised internally.
- The RAM initial block that zeroes memory is for simulation. After synthesis the power-up value depends on the target.

- The 8:1 clock ratio in TOP-003 comes from latency analysis and has not yet been simulated at the boundary (F-11).

9. Feature list for verification

The design reduces to 27 verifiable features. Each row names its source requirements, the signals to observe and a suggested method, so the table can be copied straight into a verification plan. Method key: **D** directed test, **R** constrained-random, **A** assertion, **C** functional coverage.

ID	Feature	Requirements	Observe	Method	Priority
F-01	Frame reception, mode 0, MSB first	SPI-001, SPI-002	`rx_data`	D, R	High
F-02	`rx_valid` is a one-cycle pulse after the 10th rising edge	SPI-003	`rx_valid`, `sclk`	A	High
F-03	All four `cmd` values decoded correctly	SPI-002, RAM-002 to RAM-005	`rx_data[9:8]`, RAM behaviour	D, C	High
F-04	SS_n abort after 1 to 9 bits	SPI-004, SPI-012	`rx_valid`, bit counter	D, R	High
F-05	SS_n abort during a read	SPI-004	`miso`, FSM state	D	Medium
F-06	Read shift-out: MISO changes on falling edges, MSB first, 8 bits	SPI-005, SPI-006	`miso`, `sclk`	D, A	High
F-07	MISO holds bit 0 until SS_n high, then 0	SPI-008, SPI-004	`miso`	D, A	Medium
F-08	Extra SCLK pulses after a write or address frame ignored	SPI-007	`rx_valid`, RAM state	D	Medium
F-09	Extra SCLK pulses after the 8th read bit ignored	SPI-008	`miso`	D	Medium
F-10	One frame decoded per SS_n assertion	SPI-011	`rx_valid` count	D	Medium
F-11	Clock ratio sweep: 8:1, 10:1, 20:1, 50:1	SPI-009, TOP-003	End-to-end data match	D, C	High
F-12	Reset mid-transaction	SPI-010, RAM-001	FSM state, `addr`, `miso`	D	High
F-13	Address load with `cmd` 00	RAM-002	`addr`	D	High
F-14	Write data to the loaded address	RAM-003	`mem[addr]`	D, R	High

ID	Feature	Requirements	Observe	Method	Priority
F-15	Address load with `cmd` 10 and read with `cmd` 11	RAM-004, RAM-005	`dout`, `tx_valid`, `miso`	D, R	High
F-16	Full address space walk 0x00 to 0xFF	RAM-009	Scoreboard	R, C	High
F-17	Overwrite an existing location	RAM-003	Read-back value	D	Medium
F-18	Read does not change memory or address; repeat read matches	RAM-005	`mem`, `addr`	D	Medium
F-19	Write does not auto-increment the address	RAM-003	`addr`, read-back	D	Medium
F-20	`cmd` 00 and 10 have the same effect	RAM-002, RAM-004	`addr`	D	Low
F-21	Unwritten location reads 0x00 in simulation	RAM-010	Read-back value	D	Low
F-22	`tx_valid` one cycle wide, only after `cmd` 11	RAM-007	`tx_valid`	A	High
F-23	Memory contents survive reset	RAM-001	Read-back after reset	D	Medium
F-24	Data patterns: 0x00, 0xFF, 0xAA, 0x55, walking 1 and 0	RAM-003, RAM-005	Scoreboard	D, C	Medium
F-25	Back-to-back transactions with minimum SS_n high time	SPI-004, SPI-012	Scoreboard	D, R	Medium
F-26	Read-frame dummy byte takes all 256 values	RAM-005	Read-back value	R	Low
F-27	End-to-end connectivity of the wrapper	TOP-001, TOP-002	Pins only	D	High

Suggested hierarchy probes for white-box checks and assertions: `u_spi.state`, `u_spi.rx_cnt`, `u_spi.tx_cnt`, `u_ram.addr`, `u_ram.mem`.